From a variant to a sentence somebody acts on

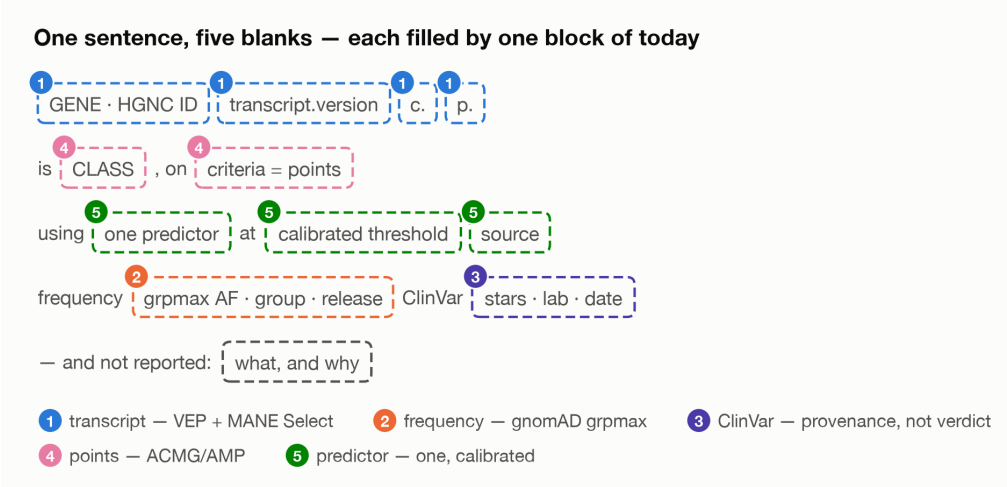
- Bir varyanttan, birinin işlem yapacağı bir cümleye
- BIB Week 13

Ad of the Week #13

- Skills Ledger: tick variant interpretation and clinical reporting

The sentence this lecture builds

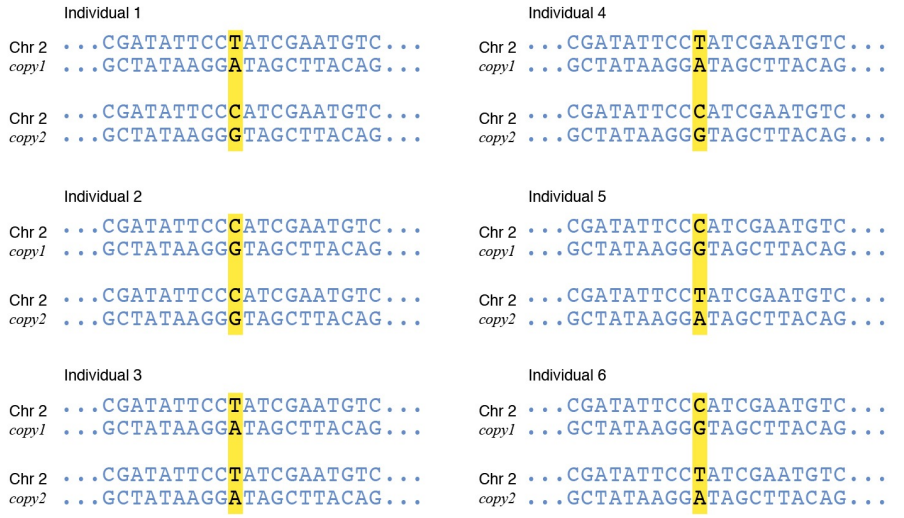
- One sentence, with a named threshold behind it:
- [variant, on a named transcript] is [class], on [criteria = points], using [one tool] at [a threshold]
- — and here is what I would not report
- Each block of today fills one blank: transcript → frequency → ClinVar → points → predictor
- This is the daily job at a licensed genetic diagnosis centre (genetik tanı merkezi)
- A classification with no named threshold behind it is an opinion



The sentence as a template: each blank is filled by one block of today's lecture

SNPs and haplotypes, in two minutes

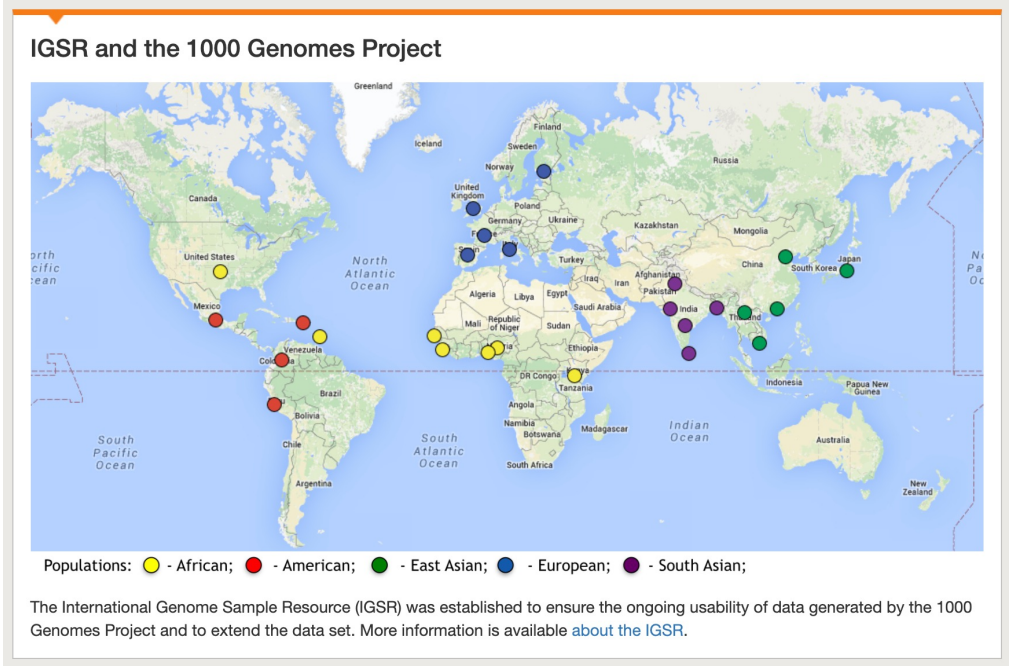
- A SNP (tek nükleotid polimorfizmi): one position where people differ; each of us carries two copies
  - same base twice: homozygous (homozigot); two different bases: heterozygous (heterozigot)
- A typical genome differs from the reference at ~4–5 million sites (1000 Genomes, Nature 2015)
- Neighbouring SNPs are inherited together in haplotype blocks (haplotip blokları)
- HapMap (2002–2010): a few tag SNPs stand in for a whole block — what a genotyping array reads
- So a GWAS hit marks a block, not a variant. Hold that for two slides.



One SNP in six people, two chromosome copies each: one C/C, one T/T, four heterozygotes (2025 deck, slide 6)

1000 Genomes: who was sampled?

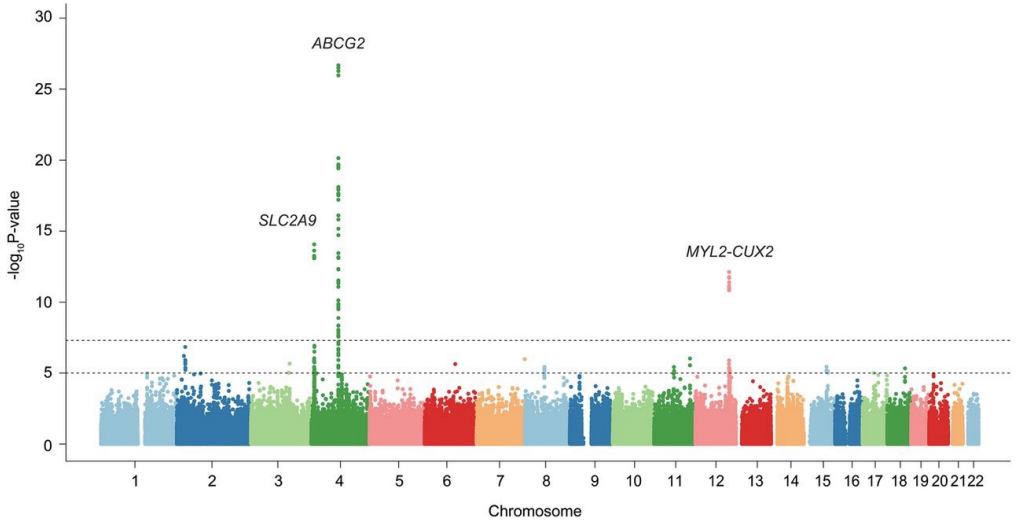
- 1000 Genomes (2008–2015): 2,504 people from 26 populations in five continental groups
- Goal: every variant at ≥1% frequency in those populations — the data now lives on as IGSR
- Find Türkiye on the map. No population from Anatolia, the Caucasus or the Middle East was sampled.
- Week 2: "the population represented in these references is not this one". This is what it meant.
- Every frequency you read today was measured on somebody. Ask on whom.



1000 Genomes / IGSR sampling sites: five continental groups, nothing between Italy and Pakistan (2025 deck, slide 8)

Where disease variation sits: a Manhattan plot, moved

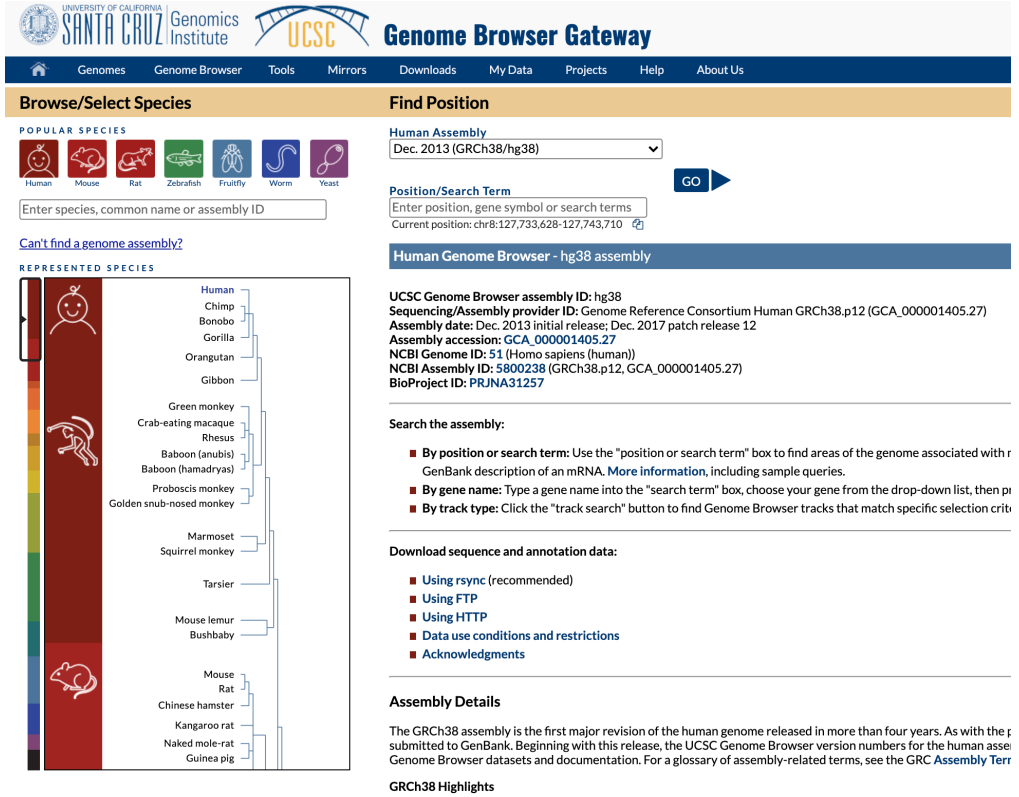
- Gout GWAS (945 cases, 1,213 controls): three loci pass $p < 5 \times 10^{-8}$ (Matsuo, Ann Rheum Dis 2016)
- Each peak is an LD block of tag SNPs — not a proven causal variant
- Most common-disease GWAS peaks fall outside coding exons
- Most GWAS heritability (kalıtsallık) sits in non-coding variation (ENCODE3, Nature 2020)
- ~10.7% of the genome is under evolutionary constraint; ~1.5% is coding (Zoonomia, Science 2023)
- Week 5 retired "97% junk". Today's missense tools speak only for the coding ~1.5%.



Gout GWAS Manhattan plot: ABCG2, SLC2A9 and MYL2–CUX2 above the genome-wide line (Matsuo et al., Ann Rheum Dis 2016)

Fix: how many genes, and in which release?

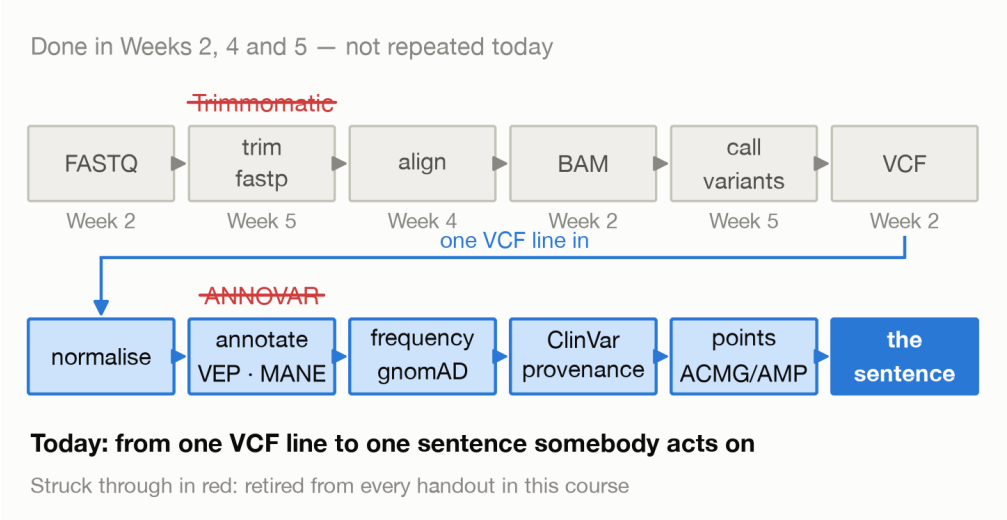
- The 2025 slide said "~25,000 genes". That number is retired.
- ~19,400–19,900 protein-coding genes (protein kodlayan gen), depending on catalogue and release
  - plus comparable numbers of lncRNA and small-RNA loci
- The count changes with every release. The genome does not.
- So a variant result names its build and its annotation release (anotasyon sürümü): GENOMES G
- Even a screenshot carries a version: this one says GRCh38.p12. Check what today's gateway says.



UCSC Genome Browser gateway (2025 capture): the page states its own assembly and patch, GRCh38.p12

SAM, BAM, CIGAR, VCF: you did this in Week 2

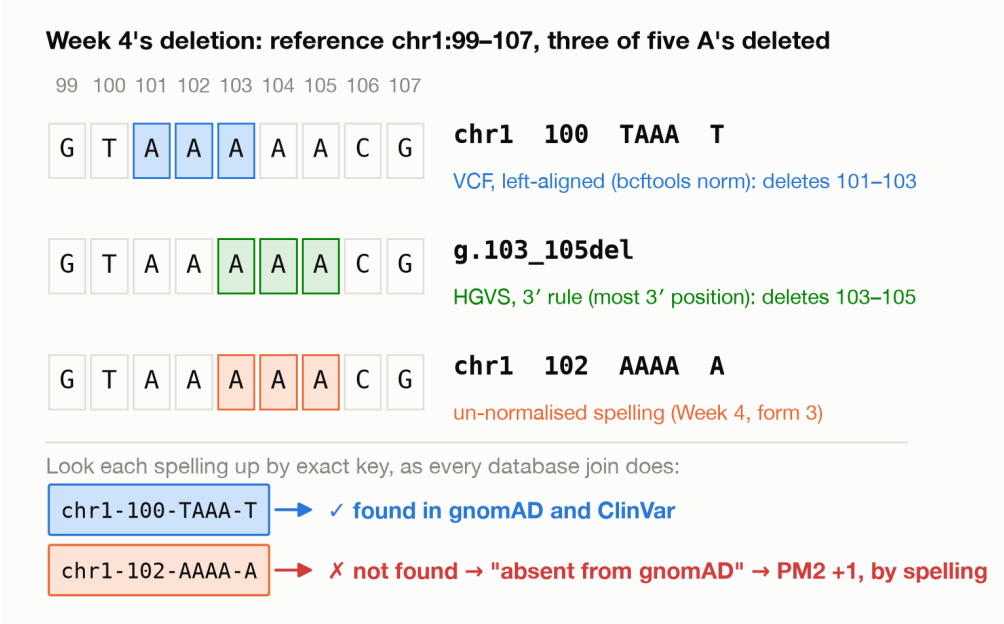
- Formats and coordinates: Week 2 · CIGAR as a traceback: Week 4 · learned variant callers: Week 5
- Today starts where your VCF ends: one variant line in, one sentence out
- Retired: ANNOVAR → Ensembl VEP or SnpEff (ANNOVAR needs registration and a licence)
- Retired: Trimmomatic → fastp (Week 5) — the 2024 semester-project handout still names both
- If an old handout disagrees with this slide, the handout is out of date



Weeks 2–5 already took you from FASTQ to VCF; today runs from one VCF line to the sentence

Before anything else: normalise the variant

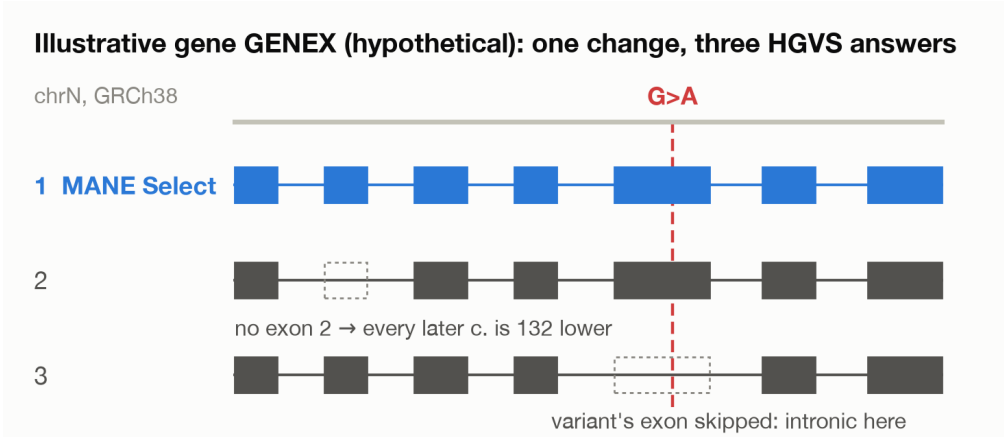
- Week 4 banked this: one deletion, four legal VCF spellings, one normalised form: chr1 100 TAAA T
- Normalise first (normalizasyon): left-align, trim, split multi-allelic sites — then look things up
- An un-normalised indel can miss its own gnomAD entry — and "absent from gnomAD" is evidence (PM2)
- HGVS shifts the same deletion the other way, to the most 3′ position: here g.103_105del
- Both are correct. Say which convention each coordinate in your report uses.
- Deposit cashed: "the first thing you will do is normalise it"



Week 4's deletion, normalised: one key that matches the databases, one HGVS name, and an un-normalised spelling that matches nothing

Which transcript? VEP, and MANE Select

- Ensembl VEP reports a consequence on every transcript (transkript) the variant overlaps
- One genomic change → different HGVS c. and p. on different transcripts — or no p. at all
- MANE v1.5 (March 2026): 19,437 transcripts over 19,367 coding genes, one MANE Select per gene
  - identical in Ensembl and RefSeq, so the report's transcript stops being a matter of taste
- Report the MANE Select line with versioned IDs: HGNC ID + ENST….N / NM_….N, never a bare symbol
- "Which transcript, which release" is part of the result — Week 2's G and Week 7's E, together

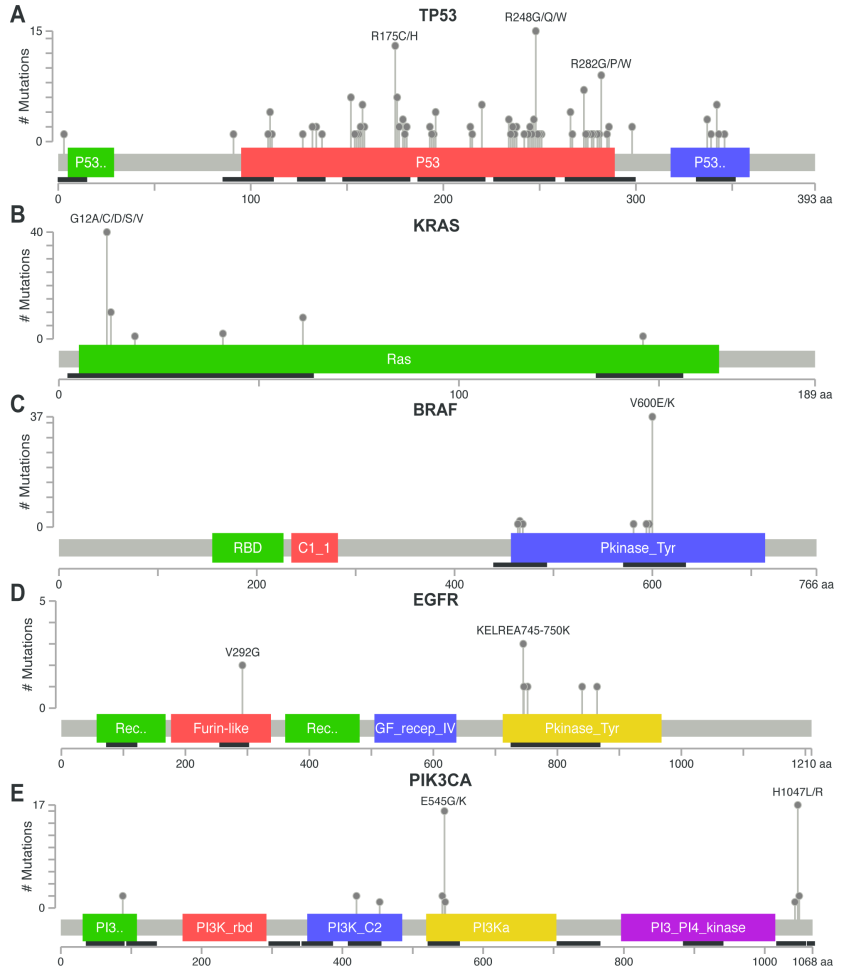


	transcript (placeholder ID)		HGVS c.	HGVS p.	consequence
1	ENST000000xxxxx.N	MANE	c.1045G>A	p.(Gly349Ser)	missense
2	ENST000000yyyyy.N		c.913G>A	p.(Gly305Ser)	missense
3	ENST000000zzzzz.N		c.880+412G>A	–	intron

Illustrative gene GENEX: one genomic G>A, three transcripts, three different HGVS answers

Where on the protein? The lollipop plot

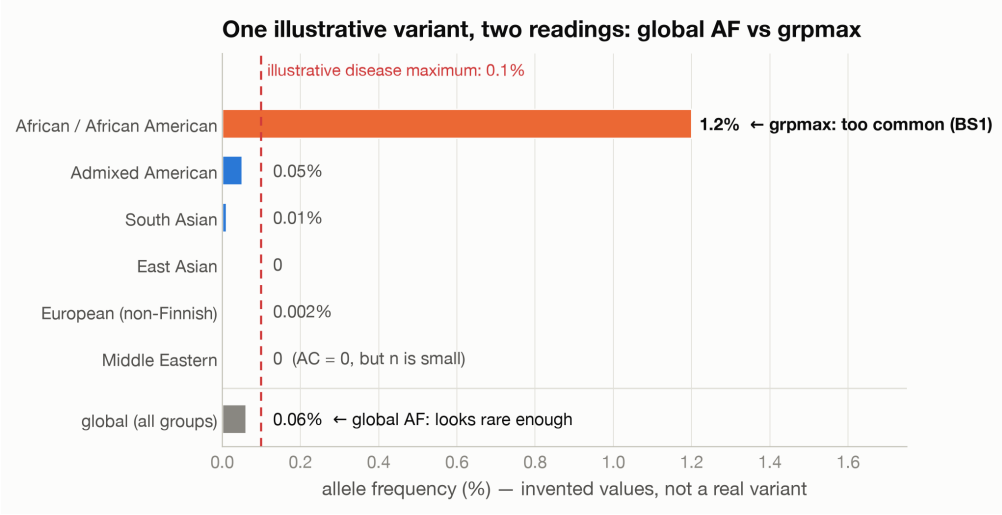
- With the p. position fixed on MANE Select, a lollipop plot (lolipop grafiği) shows where variants fall
- x = amino-acid position · stick height = times observed · coloured blocks = protein domains
- Hotspots cluster in functional domains: TP53 R175 and R248, KRAS G12, BRAF V600E
- A hotspot or a critical domain is ACMG evidence in its own right (PM1)
- Change the transcript and every lollipop moves — the previous slide, drawn
- These are somatic tumour variants; today's points system is for germline variants



Lollipop plots of somatic mutations in TP53, KRAS, BRAF, EGFR and PIK3CA (PLOS ONE 2016; 2025 deck, slide 20)

Frequency evidence: read grpmax, not global AF

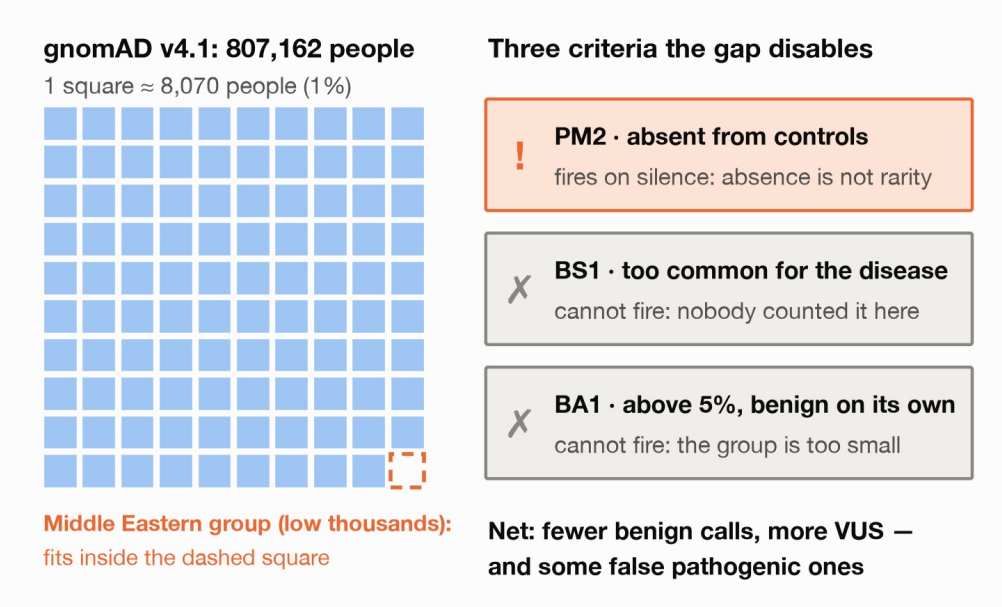
- gnomAD v4.1: 807,162 individuals — 730,947 exomes + 76,215 genomes
- Global AF (alel frekansı) averages over everyone, so the largest groups dominate it
- grpmax: the highest frequency in any of the main genetic ancestry groups — read this one
- A variant at 1% in one group can look "rare" globally — rare enough to be called pathogenic
- The only question frequency answers: is this too common to cause this disease?



Illustrative variant: 1.2% in one group, 0.06% globally — global AF says 'rare', grpmax says 'too common'

The Middle Eastern gap: three criteria switch off

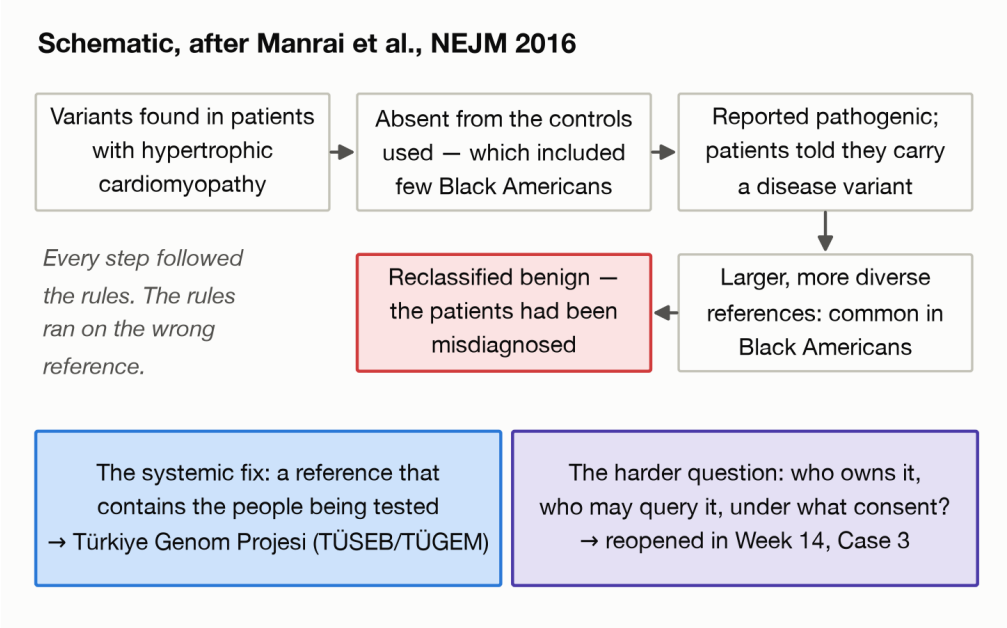
- gnomAD's Middle Eastern group is in the low thousands — under 1% of 807,162 people
- PM2, BS1 and BA1 are the criteria that missing frequency data disables
  - PM2 "absent from controls" fires on absence of evidence; BS1 and BA1 cannot fire on data never collected
- So a Turkish patient's variant is likelier to come back a VUS — or wrongly pathogenic
- Consanguinity (akraba evliliği) shifts it further: more homozygous rare variants, more founder alleles
- Before you read any frequency, ask: measured on whom, and how many?



Left: gnomAD v4.1 as 100 squares of ~8,070 people — the Middle Eastern group fits inside one. Right: the three frequency criteria the gap disables

Ethics beat: when the reference was somebody else

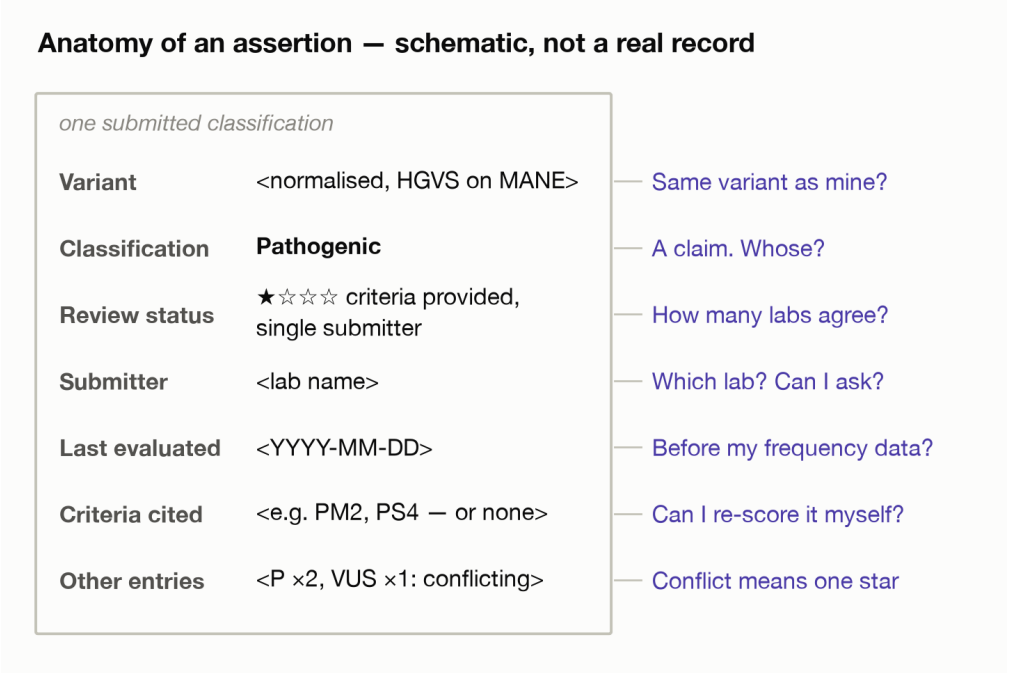
- Manrai et al., NEJM 2016: hypertrophic-cardiomyopathy variants had been reported pathogenic
- They were benign polymorphisms, common in Black Americans — and patients were misdiagnosed
- Mechanism: "absent from controls" — when the controls did not include them
- The systemic fix is a population reference for Türkiye: TÜSEB/TÜGEM, the Türkiye Genom Projesi
- The harder question — who owns it, who may query it, under what consent — is Week 14's



Schematic, after Manrai et al. NEJM 2016: absent from the controls → reported pathogenic → common in a population nobody had counted → reclassified benign

ClinVar: somebody else's classification, with a date

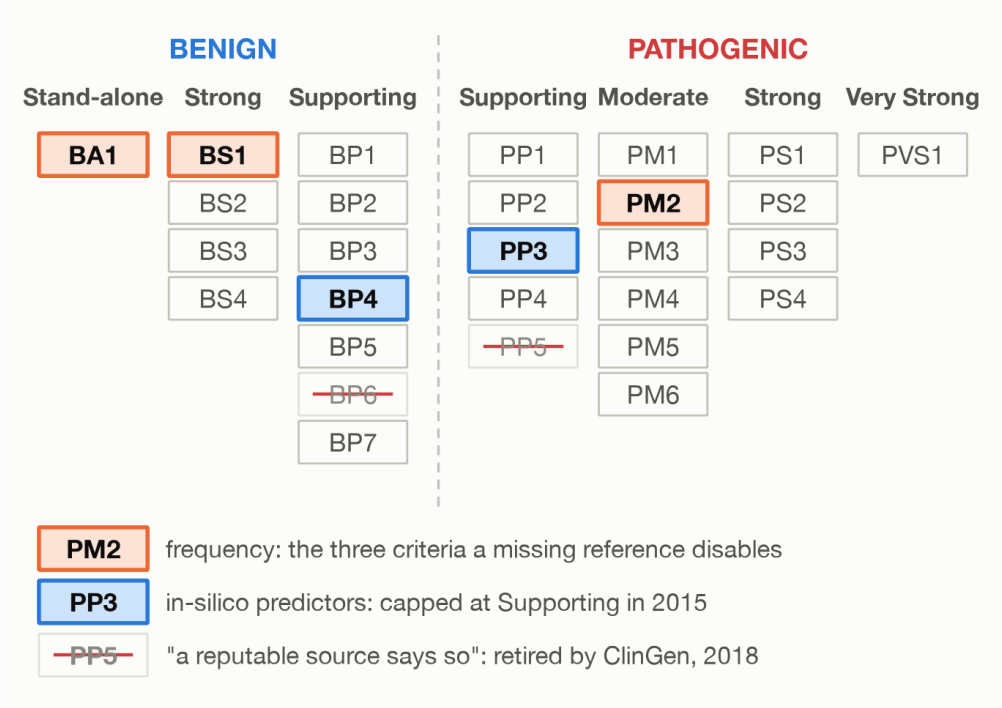
- ClinVar collects classifications submitted by labs. It does not classify anything itself.
- Read four things: review status (inceleme durumu), submitter, last-evaluated date, conflicts
- Stars: none = no criteria · ★ one lab, or conflicting · ★★ labs agree · ★★★ panel · ★★★★ guideline
- A classification last evaluated years ago predates the frequency data you are using — re-score it
- "A reputable source says pathogenic" used to be a criterion (PP5); ClinGen retired PP5 and BP6 in 2018



Anatomy of an assertion: the fields of a ClinVar submission and the question each one answers (schematic — not a real record)

ACMG/AMP 2015: 28 criteria, four strengths

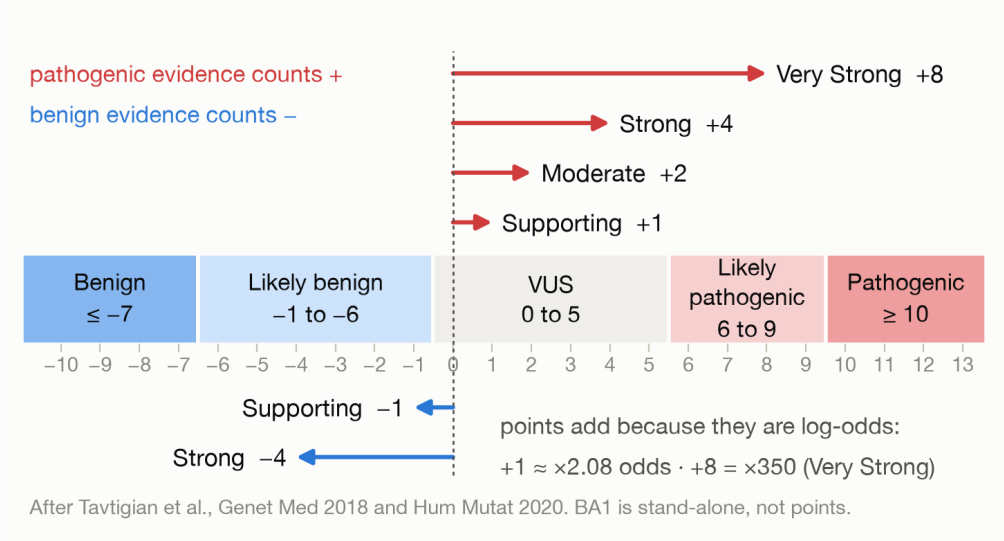
- Richards et al., Genet Med 2015: 28 criteria (ölçüt) — 16 pathogenic, 12 benign
- Each has a strength: Supporting, Moderate, Strong, Very Strong (benign: up to Stand-alone)
- Five classes: Pathogenic · Likely Pathogenic · VUS (önemi belirsiz varyant) · Likely Benign · Benign
- "Likely" means roughly >90% certain. A VUS is not a weak pathogenic — it means unknown.
- Computational evidence (PP3/BP4) was capped at Supporting; no tool or threshold was endorsed



The 28 criteria at their default strengths: frequency criteria in orange, in-silico predictors in blue; PP5 and BP6 retired in 2018

Points: the same rules, as arithmetic

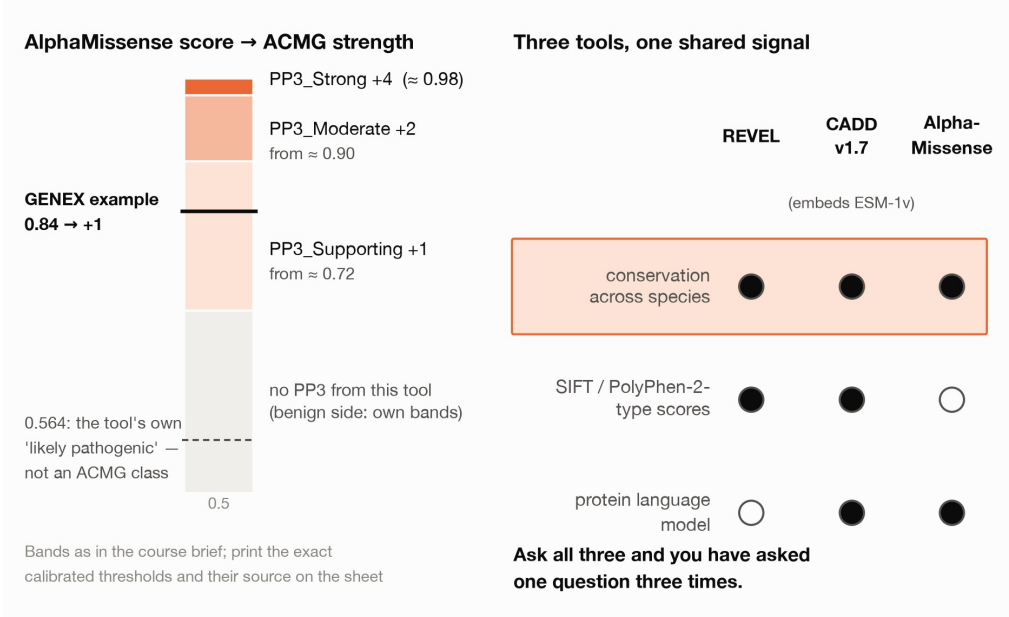
- Tavtigian et al. recast the rules as Bayesian odds in 2018, then as points (puan) in 2020
- Supporting ±1 · Moderate ±2 · Strong ±4 · Very Strong ±8 — pathogenic +, benign −
- Add them up: P ≥ 10 · LP 6 to 9 · VUS 0 to 5 · LB −1 to −6 · B ≤ −7
- Points add because they are logarithms: each point multiplies the odds of pathogenicity by about 2
- So "how far is this VUS from Likely Pathogenic?" now has an answer, in points



The points line: evidence strengths as steps of 1, 2, 4 and 8; class boundaries at −7, −1, 6 and 10

One calibrated predictor, at a named threshold

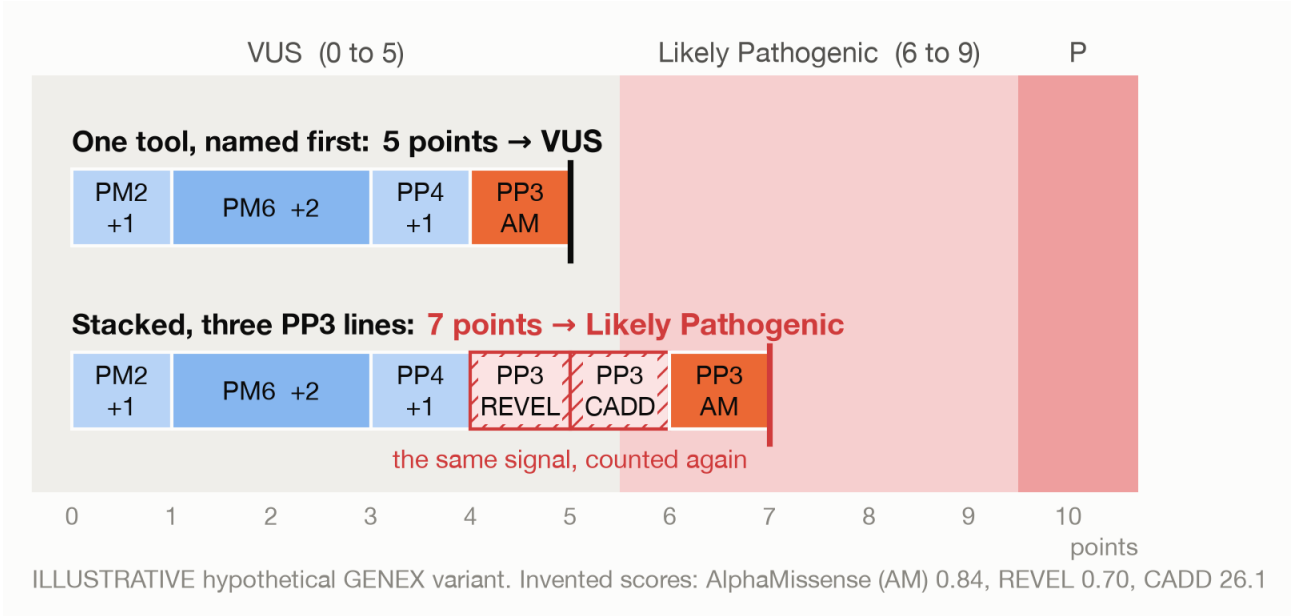
- Pejaver et al., AJHG 2022 (ClinGen): 13 missense predictors calibrated to strengths, up to Strong
- AlphaMissense, calibrated later the same way: ≈0.72 Supporting · ≈0.90 Moderate · ≈0.98 Strong
- Rule one, the one-tool rule: one calibrated predictor per variant, chosen before you see scores
  - REVEL, CADD and AlphaMissense share data and features: stacking counts one signal several times
  - "Classical scores vs deep learning" is a false split: CADD v1.7 already embeds ESM-1v (NAR 2024)
- Rule two: PP3/BP4 is never sufficient alone — a prediction can tip the evidence, never be it



Left: AlphaMissense score bands as ACMG strengths (≈0.72 / 0.90 / 0.98). Right: why three tools are not three witnesses

Worked example: one variant, scored twice

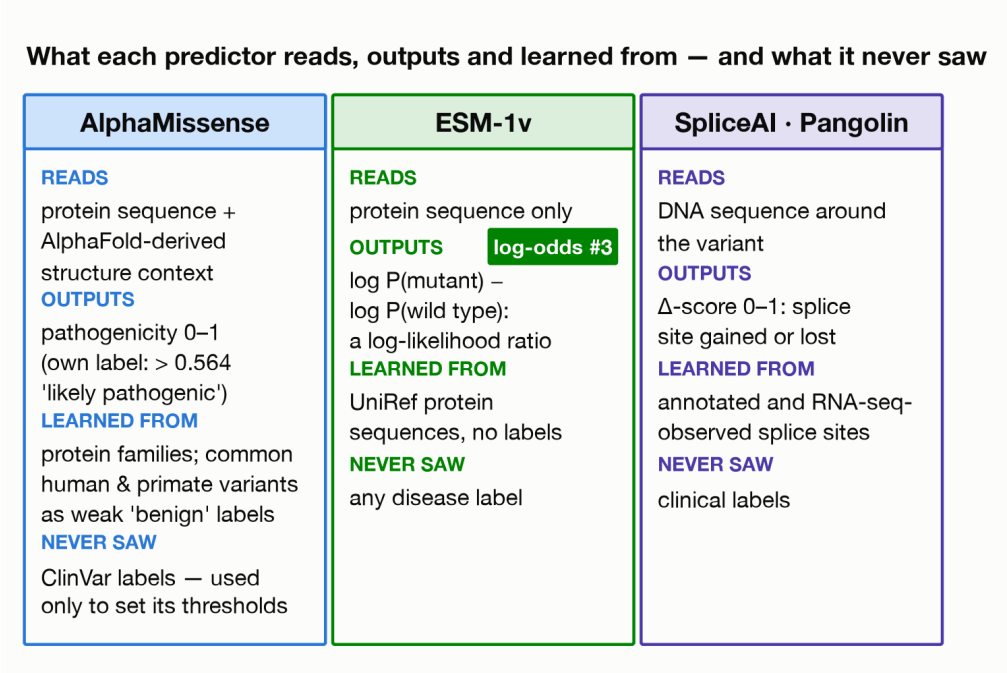
- Illustrative GENEX missense: de novo (parents unconfirmed), absent from gnomAD
- Shared: PM2_Supporting +1 · PM6 +2 · PP4 +1 = 4
- One tool, named first: AlphaMissense 0.84 → PP3 +1 → 5 = VUS
- Stacked: REVEL, CADD, AlphaMissense as three PP3 lines → +3 → 7 = LP
- Nothing new was learned: one signal, counted three times



Illustrative GENEX variant: 5 points (VUS) with one calibrated predictor; 7 points (Likely Pathogenic) when three predictors are stacked

What the predictors actually are

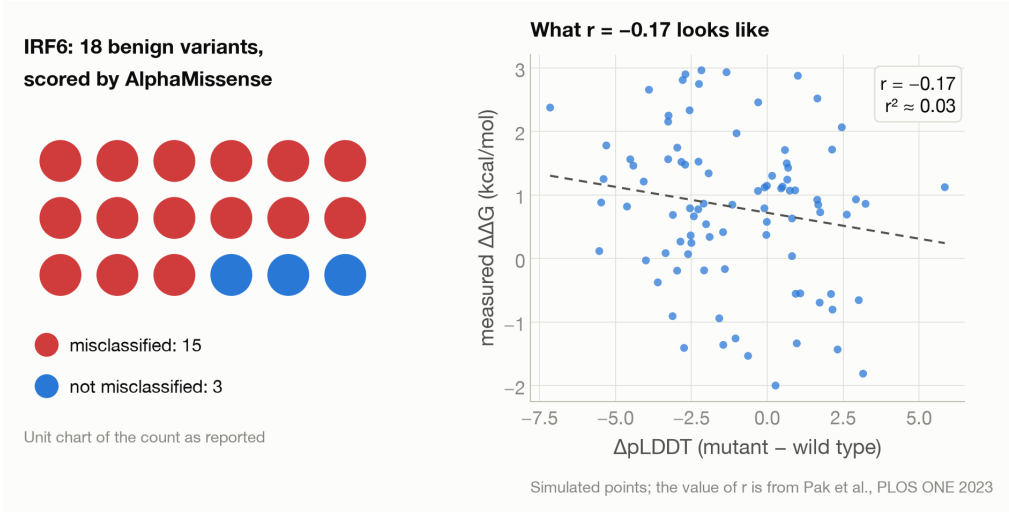
- AlphaMissense: AlphaFold representations + population frequency as weak labels, not ClinVar labels
- ESM-1v: score = log P(mutant) − log P(wild type) at that position — a log-likelihood ratio
  - Log-odds, third meeting: BLOSUM62 (Week 4) → model log-likelihoods (Week 11) → today
- SpliceAI and Pangolin predict splice changes from DNA: the top non-coding tools in diagnostics
- AlphaMissense's own "likely pathogenic" label (> 0.564) is not ACMG Likely Pathogenic
- Ask each one Week 4's question: what was the observed set?



What each predictor reads, outputs and learned from — and what it never saw

Who benchmarked it, and on which gene?

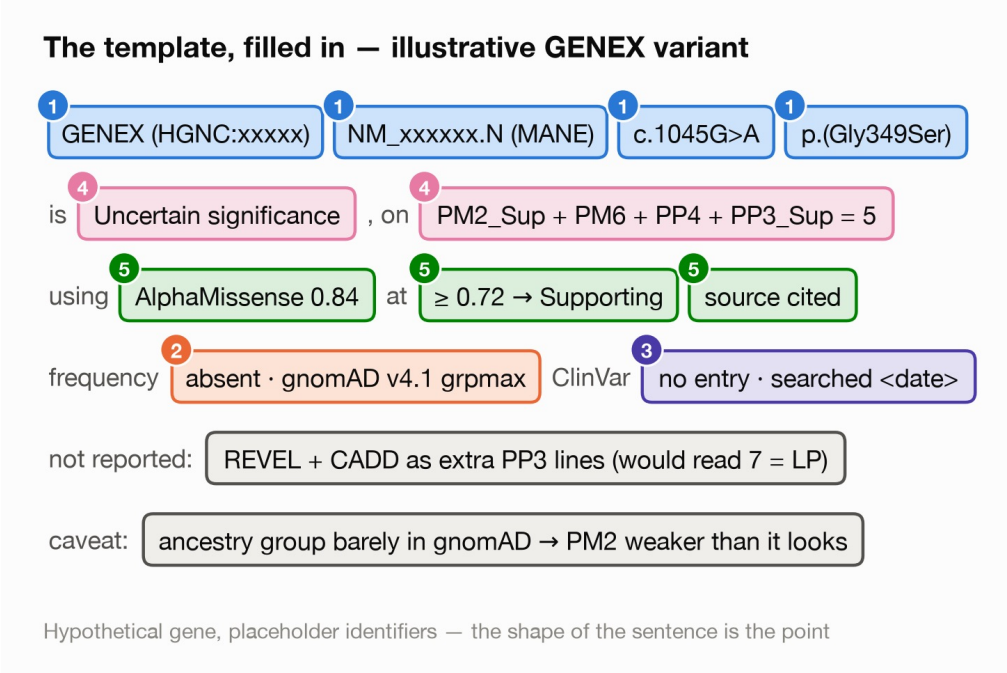
- Benchmark provenance (Week 11): PrimateAI-3D beats AlphaMissense on four real cohorts
  - …in an Illumina-authored preprint; Illumina makes PrimateAI-3D. Report the result and the interest.
- Calibration is not accuracy: AlphaMissense misclassified 15 of 18 benign IRF6 variants
  - A global AUROC says nothing about your gene
- Week 12, cashed: ΔpLDDT vs measured ΔΔG, r ≈ −0.17 (Pak et al., PLOS ONE 2023)
- pLDDT is not a mutation-effect predictor: variant questions come here, to a calibrated tool



Left: 15 of 18 benign IRF6 variants misclassified. Right: what r = −0.17 looks like (simulated data)

The sentence, filled in

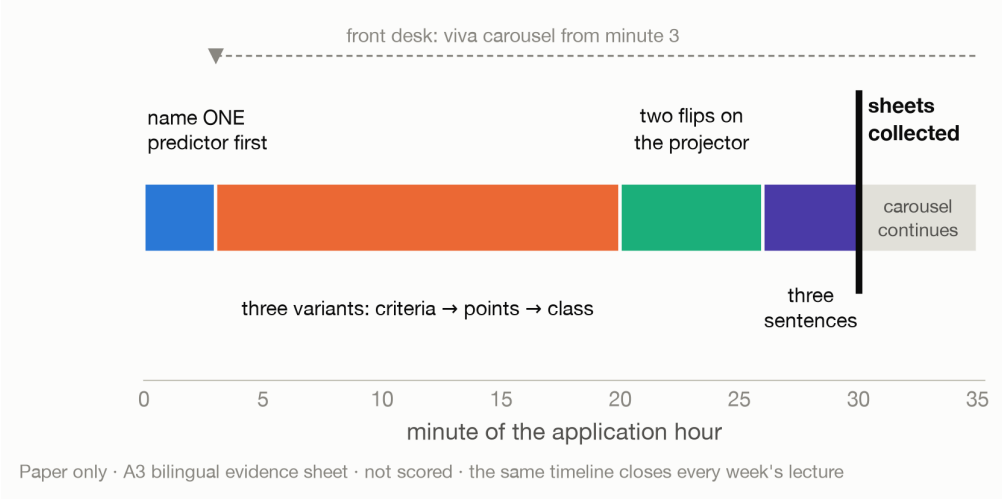
- Illustrative: GENEX (HGNC:xxxxx) NM_xxxxxx.N (MANE Select) c.1045G>A p.(Gly349Ser), heterozygous
- Uncertain significance, 5 points: PM2_Supporting, PM6, PP4, PP3_Supporting
- PP3 from AlphaMissense 0.84 at the Supporting threshold (≈0.72), calibration source named
- Not reported: REVEL and CADD as extra PP3 lines — 7 points, LP, by counting one signal three times
- Caveat: the patient's ancestry group is barely in gnomAD, so "absent" is weaker than it looks



The template from minute 4, filled in for the illustrative GENEX variant — every clause carries a source or a version

This afternoon: three variants, three sentences, on paper

- Application hour, 35 minutes, paper only — no laptop, no network
- A3 bilingual evidence sheet: one clear pathogenic, one clear benign, one predictor-disagreement trap
- First, at the top: your ONE calibrated predictor, its calibration source, its exact threshold
- Then criteria → points → class → the sentence. Two flips are shown once on the projector.
- Sheets collected at minute 30. Not scored. The viva carousel runs at the front desk from minute 3.



The application hour: name your predictor, score three variants, watch two flips, write three sentences — sheets collected at minute 30